

Multi-Source Candidate Data Transformer — Technical Design

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Problem framing

Many noisy sources (CSV, ATS JSON, resumes, recruiter notes) describe the same candidate in conflicting ways. Downstream products need **one** canonical profile per person with normalized formats, deduped values, a record of where each value came from (provenance), and a calibrated confidence. The cost of a wrong-but-confident value is high: it silently pollutes hiring decisions. Honest-empty > wrong-confident.

Pipeline

detect → **extract** → **normalize** → **merge** → **confidence** → **project-to-output** → **validate**. Each stage is a pure function over the previous stage's output, which makes the run deterministic and easy to test. The internal canonical record is the single source of truth; projection is a thin layer that *reshapes* but never *mutates* it.

Canonical schema & normalized formats

- **phones**: E.164 (e.g. +14155552671); region inferred from country, else US default.
- **dates**: YYYY-MM. "present"/"now" → null on the *end* field.
- **country**: ISO-3166 alpha-2 via small alias map (USA→US, UK→GB, India→IN, ...).
- **skills**: dictionary-based canonical names (js→JavaScript, k8s→Kubernetes, ...). Unknown skills pass through verbatim (no invented canonicalization).
- **emails**: lower-cased, regex-validated; bad strings drop to null.

Merge & conflict-resolution policy

Identity is resolved by **union-find** over three match keys (in order): *normalized email* → *E.164 phone* → *normalized full name*. Conflicting scalars (full_name, headline, location.*) are won by **source priority** (*ats_json=5, recruiter_csv=4, resume=3, notes=2*); ties → first non-empty. List fields (emails, phones, education, experience) are unioned and deduped by stable keys. Skills are merged by canonical name; corroboration boosts confidence with diminishing returns: $c' = 1 - (1 - c_1)(1 - c_2)$, capped at 0.99. No source is allowed to invent a value missing from its own input.

Confidence

Per-field score from the top source's base (priority→base map) plus corroboration boost per additional source. **overall_confidence** is a weighted mean over present fields only — empty fields contribute zero weight and don't inflate the score. *Provenance is a first-class output*: every populated field carries (field, source, method) so reviewers can audit any number.

Runtime custom-output config

A JSON config describes the desired shape: subset of fields, renames via *from* (supports dotted paths, *[N]* indexing, *[]* flattening), per-field *normalize* (E164, canonical) and type coercion, plus *on_missing* ∈ {null, omit, error} and toggles for provenance / confidence. Projection is validated against the requested shape; the canonical record is untouched, so two configs against the same inputs are guaranteed consistent.

```
{ "fields": [ { "path": "primary_email", "from": "emails[0]", "required": true },
               { "path": "phone", "from": "phones[0]", "normalize": "E164" },
               { "path": "skills", "from": "skills[].name", "normalize": "canonical" } ],
  "include_confidence": true, "on_missing": "null" }
```

Edge cases — what I handle, what I cut

- **Same person across sources, different phone formats** → normalized to E.164 before identity match; one record out.
- **Conflicting full_name ("Jane Doe" vs "Jane D.")** → matched on shared email; ATS wins by priority.
- **Garbage row** (country "Atlantis", phone "abc") → fields drop to null, profile still emitted; provenance shows nothing was promoted.
- **Missing source file** → warning logged, pipeline continues with remaining sources.
- **No identity signal** (recruiter notes with no email/phone) → kept as its own record rather than guessing identity (false-merge would be worse than a duplicate).
- **Descope under time pressure**: live GitHub/LinkedIn fetching, PDF/DOCX → text conversion, fuzzy name matching. Modules are stubbed for easy add-on.

Constraints met

Deterministic & explainable: same inputs → same output (verified by test). Every value is traceable via provenance. **Robust**: missing/garbage sources never crash the run; unknown values stay null. **Scale**: pure-Python pipeline is O(N) per source; identity uses union-find — comfortable on thousands of candidates on a laptop. **Surface**: a thin CLI (*python -m src.cli --csv ... --ats ... --config ... --out ...*).